TS1000
TS2000
TS2100
Robot Controller

INSTRUCTION MANUAL

TS1000/TS2000 ROBOT CONTROLLER

USER PARAMETER MANUAL

Notice

- Make sure that this instruction manual is delivered to the final user of Toshiba Machine's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

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TOKYO, JAPAN

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Preface

This manual describes the machine parameters for small-size SCARA robot controller "TS1000/TS2000".
The machine parameters are registered in the controller under the text file named "USER.PAR". Like the SCOL program, they can be edited by the program editor.
For use of the program editor, see the Operator's Manual.

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[U00] Robot type

Specify the type of the robot.

```
[U00] Robot type
= "SCARA"
```

For the small-size SCARA robot, specify "SCARA".

[U01] System BASE coordinate

Specify the values of the base coordinate system, which become effective at power ON.

```
[U01] System BASE coordinate
= 0.0   0.0   0.0   0.0   0.0   0.0
```

Specify the values in the order of X, Y, Z, C, 0.0, 0.0.
To make the base coordinate system identical with the world coordinate system, specify "0.0 0.0 0.0 0.0 0.0 0.0".

[U02] NOWAIT mode for multi task

Specify the multitask operation mode.

```
[U02] NOWAIT mode for multi task (0: Common, 1: Individual)
= 0
```

Specify whether system variable "NOWAIT" should be controlled independently for each task or commonly used by tasks during multitask operation.
If the common use by tasks is selected, however, the SCOL programming becomes complicated. We recommend to use the independent setting for each task.

Set values =0 Independent control for each task (NOWAIT improved type) (Toshiba Machine's recommendation)
=1 Common use by tasks (NOWAIT previous type)

[U03] Default MOVESYNC mode

Specify the synchronous or asynchronous mode for a motion command (i.e., initial value of MOVESYNC).

[U03] Default MOVESYNC mode (0: ENABLE, 1: DISABLE) = 1

When the motion command synchronous mode is selected, the system executes just before the next motion command and waits there for the finish of positioning during programmed operation.
When the motion command asynchronous mode is chosen, the system pre-reads and executes a maximum of four (4) motion commands.
In the motion command asynchronous mode, pass motion is not possible even if "PASS" is set to ENABLE.

Set values =0 Motion command synchronous mode (ENABLE MOVESYNC)
=1 Motion command asynchronous mode (DISABLE MOVESYNC)

[U04] TEACHING mode override

Specify the maximum speed of a motion in the teaching mode.

[U04] TEACHING mode override = 20

Specify the value in the range of 1 ~ 100 %.

When the set value is 20, for instance, it signifies that the speed is limited to 20 % of the maximum speed.

[U05] Override on slow speed signal

Specify the speed at the command of slow speed (system input signal).

```
[U05] Override on slow speed signal (0 – 100) [%]
= 25
```

Specify the robot operating speed at the input of a slow speed command signal. If the slow speed command signal is input during automatic operation, the robot operates at a speed which is the product of the maximum speed and this parameter value (override). The set range is 1 ~ 100 %.

When the set value is 25, for instance, it signifies that the speed is limited to 25 % of the maximum speed.

[U06] Serial port setting

Specify the communication port (i.e., COM port).

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```
[U06] Serial port setting
  {Speed} (38400, 19200, 9600, 4800, 2400, 1200)
  {Character} (7, 8)
  {Parity} (0: Without, 1: Odd, 2: Even)
  {Stop bit} (1, 2)
= 9600   8   0   1
= 38400   8   0   1
```

= (Speed) (Character length) (Parity) (Stop bit length) ← Setting of COM1 port

= (Speed) (Character length) (Parity) (Stop bit length) ← Setting of HOST port

TS2000

```
[U06] Serial port setting
  {Speed} (38400, 19200, 9600, 4800, 2400, 1200)
  {Character} (7, 8)
  {Parity} (0: Without, 1: Odd, 2: Even)
  {Stop bit} (1, 2)
  [COM1]
= 9600   8   0   1
  [HOST]
= 9600   8   0   1
  [COM2]
= 9600   8   0   1
```

[COM1]

= (Speed) (Character length) (Parity) (Stop bit length) ← Setting of COM1 port

[HOST]

= (Speed) (Character length) (Parity) (Stop bit length) ← Setting of HOST port

[COM2]

= (Speed) (Character length) (Parity) (Stop bit length) ← Setting of COM2 port

(Speed): Specify the data communication speed. Select from the following six (6) data.

38400	:	38400 bps
19200	:	19200 bps
9600	:	9600 bps
4800	:	4800 bps
2400	:	2400 bps
1200	:	1200 bps

(Character length): Specify the character length to be transferred.

8 : 8 bits

7 : 7 bits

(Parity): Specify the character parity to be transferred.

0 : Without parity
1 : Odd parity
2 : Even parity

(Stop bit length): Specify the stop bit length of characters to be transferred.

1 : Stop bit 1
2 : Stop bit 2

Example: TS1000

Set value = 9600 8 0 1
= 38400 8 1 1

Setting of COM1 port: Speed 9,600 bps, character length 8 bits, without parity and stop bit 1.

Setting of HOST port: Speed 38,400 bps, character length 8 bits, even parity and stop bit 1.

[U07] Specify signal for EXTSELECT

Setting of file select signal line.

```
[U07] Specify signal for EXTSELECT
  {Signal No.} (1–  )
  {Bit length} (1 – 4)
= 101  4
```

Specify the input signal from outside the controller to select an execution file, using the input signal.
Specify up to four (4) consecutive input signals from standard inputs 1 ~ 16, extended inputs 101 ~ 164 or field bus inputs 301 ~ 364. To be more specific, specify the leading number of input signals to be used and the number of signals to be used by discriminating them by space.

= (Signal No.) (Bit length)

(Signal No.): Specify the leading number of input signals to be used.
Set value 1 ~ 64 (Standard input)
101 ~ 164 (Extended input)
301 ~ 364 (Field bus input)
Select the number from the above inputs.

(Bit length): Specify the number of signals to be used.
Set value TS1000: 1 ~ 4
TS2000: 1 ~ 8

Example: Set value = 101 4

Four (4) extended input signals numbered 101 ~ 104 are used.

[U08] EXTSELECT file (TS1000 only)

Specify the program files corresponding to file select signal lines.
In the TS2000 controller, they are set in the exclusive file "EXTRNSEL.SYS".
For details, see the descriptions on "EXTRNSEL.SYS" carried in this manual.

```
[U08] EXTSELECT file
= "PROG1"
= "PROG2"
= "PROG3"
= "PROG4"
= "PROG5"
= "PROG6"
= "PROG7"
= "PROG8"
= "PROG9"
= "PROG10"
= "PROG11"
= "PROG12"
= "PROG13"
= "PROG14"
= "PROG15"
= "PROG16"
```

The number of effective files differs with the bit length as designated in [U07] above, and the program files are assigned in turn, starting with the top file.

Bit length 1: Two (2) files of 0 ~ 1 (binary numbers).
Bit length 2: Four (4) files of 00 ~ 11 (binary numbers).
Bit length 3: Eight (8) files of 000 ~ 111 (binary numbers).
Bit length 4: Sixteen (16) files of 0000 ~ 1111 (binary numbers).

Example: Set value = "PROG1"

PROG1 is assigned to the file select signal.

[U09] Auxiliary signal

Specify the data on the auxiliary signal display so that the robot I/O can be set ON and OFF from the teach pendant.

```
[U09] Auxiliary signal
  {Type }(0:Single, 1:Double)
  {Signal No. }(1 - 24)
  {Signal Name}(Max 10character)
= 0   201   "Hand out 1"
= 0   202   "Hand out 2"
= 0   203   "Hand out 3"
= 0   204   "Hand out 4"
= 0     0   "Not   Use   "
= 0     1   "Dout 1      "
= 0     2   "Dout 2      "
= 0     3   "Dout 3      "
= 0     4   "Dout 4      "
= 0     5   "Dout 5      "
= 0     6   "Dout 6      "
= 0     7   "Dout 7      "
= 0     8   "Dout 8      "
= 0     9   "Dout 9      "
= 0    10   "Dout 10     "
= 0    11   "Dout 11     "
= 0    12   "Dout 12     "
= 0    13   "Dout 13     "
= 0    14   "Dout 14     "
= 0    15   "Dout 15     "
```

= (Type) (Signal No.) (Signal name)

Output signals for twenty (20) contacts can be specified.

(Type): When the double solenoid is connected with output signals, the output signals should be controlled exclusively. When "Double" is specified, two (2) successive output signals are controlled exclusively.
0 : Single solenoid (normal output)

1 : Double solenoid (Two (2) consecutive output signals are output exclusively.)

Caution: This is the setting for auxiliary signal operation through the teach pendant. Even if the double solenoid is selected, signals are not output exclusively at automatic operation. Exclusive output of signals should be specified in the program.
Even if the double solenoid is specified, all output signals are OFF at power ON.

(Signal No.): Specify the number of output signal to be used.
When "Double" is specified, specify the smaller number of two (2) consecutive output signals. Output signals that can be specified are as follows:

0	Not registered.				
1	(DOUT1)	~	16	(DOUT16)	Standard output
101	(DOUT101)	~	164	(DOUT164)	Extended digital output
201	(HANDOUT1)	~	214	(HANDOUT4)	Hand output

(Signal name): Specify the signal name displayed. Up to ten (10) characters can be set.

Example 1: Set value = 0 201 "Hand out 1"

Output signal 201 is registered as the single solenoid and the signal name is "Hand out 1".

Example 2: Set value = 1 1 "out 1"

Output signals 1 and 2 are registered as the double solenoid and the signal name is "out 1".

[U10] Default Teaching mode

Specify the initial value of manual operation in the teaching mode.

```
[U10] Default Teaching mode
  {Coordinate  =  0: Joint, 1: Tool, 2: Work, 3: World}
  {Speed       =  0: Low, 1: Mid, 2: High}
  {Jog mode    =  0: Jog, 1: Inching, 2: Free}
= 3 0 0
```

Example: Set value = 3 0 0 (World, Low, Jog)

Specify the data in the order of (Coordinate) (Speed) (Jog mode)

(Coordinate): Specify the guidance coordinate system selected as default.
- 0 : Joint coordinate system
- 1 : Tool coordinate system
- 2 : Work coordinate system
- 3 : World coordinate system

(Speed): Specify the guide rate selected as default.
- 0 : Low
- 1 : Mid
- 2 : High

(Jog mode): Specify the guidance mode selected as default.
- 0 : Jog
- 1 : Inching
- 2 : Free

[U11] I/O mode

Specify the I/O operation mode.

```
[U11] I/O mode
  {Default/User} (0: Default, 1: User)
= 0
```

In the TS1000/TS2000, I/Os specified in the program are calculated by PLC, which are then input or output. This parameter selects the storage area of such sequence program.

Set values	=	0	Standard sequence
	=	1	User RAM (without debug backup)
	=	2	User flash memory
	=	3	User battery backup RAM (TS2000 only)

TS2000 allows to back up the RAM area also by battery.
Specify "1" for the parameter, which should be changed to "3" after the sequence debug. (Value 1 or 3 is saved in the same area. When "1" is specified, the sequence is cleared by power OFF/ON.)

[U12] Extend I/O setting

Set the extended I/Os.

```
[U12] Extend I/O setting
  {Use/Not Use} (0: Not use, 1: Use)
  {Input} (0 – 32)
  {Output} (0 – 32)
= 0   0   0
= 0   0   0
```

= (Use/Not Use) (Reserved) (Reserved) ← Setting of distribution I/O node 0.

= (Use/Not Use) (Reserved) (Reserved) ← Setting of distribution I/O node 1.

For the extended I/Os, two (2) nodes (node 1 and node 2) can be connected.
Node 0 is set in the upper level, and node 1 in the lower level.

(Use/Not use): Specify the presence or absence of extended I/O.
- 0 : Absence of extended I/O
- 1 : Presence of extended I/O

(Reserved): Reserved for future system extension. Specify zero (0).

(Reserved): Reserved for future system extension. Specify zero (0).

[U13] Select input signal

Specify the system inputs which are made valid.

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[U13] Select input signal (Default I/O mode only) {input 9 – 16} (0: System, 1: User) = 0 0 0 0 0 0 0 0

When "Default (standard sequence)" is specified for [U11], inputs 9 ~ 16 can be set as the user input signals or system input signals.

Specify the data in the following order.
= (input9) (input10) (input11) (input12) (input13) (input14) (input15) (input16)

Input signal line	0: System	1: User
INPUT9	STROBE	DIN9
INPUT10	PRG_RST	DIN10
INPUT11	STEP_RST	DIN11
INPUT12	CYC_RST	DIN12
INPUT13	DO_RST	DIN13
INPUT14	ALM_RST	DIN14
INPUT15	RUN	DIN15
INPUT16	EX_SVON	DIN16

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[U13] Select input signal (Default I/O mode only) {input 23, 24, 33 – 38} (0: System, 1: User) = 1 1 0 0 0 0 0 0

When "Default (standard sequence)" is specified for [U11], inputs 23, 24, 33 ~ 38 can be set as the user input signals or system input signals.

Specify the data in the following order.
= (input23) (input24) (input33) (input34) (input35) (input36) (input37) (input38)

Input signal line	0: System	1: User
INPUT23	Reserved (for extending function)	DIN23
INPUT24	ALM_RST	DIN24
INPUT33	STROBE	DIN33
INPUT34	PRG_RST	DIN34
INPUT35	STEP_RST	DIN35
INPUT36	CYC_RST	DIN36
INPUT37	DO_RST	DIN37
INPUT38	RUN	DIN38

[U14] Select output signal

Specify the system outputs which are made valid.

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```
[U14] Select output signal (Default I/O mode only)
  {output 9 – 16} (0: System, 1: User)
= 0   0   0   0   0   0   0   0
```

When "Default (standard sequence)" is specified for [U11], outputs 9 ~ 16 can be set as the user output signals or system output signals.

Specify the data in the following order.

= (output9) (output10) (output11) (output12) (output13) (output14) (output 15) (output16)

Output signal line	0: System	1: User
OUTPUT9	ACK	DOUT9
OUTPUT10	SV_RDY	DOUT10
OUTPUT11	EXTSIG	DOUT11
OUTPUT12	READY	DOUT12
OUTPUT13	AUTORUN	DOUT13
OUTPUT14	CYC_END	DOUT14
OUTPUT15	LOW_ST	DOUT15
OUTPUT16	BT_ALM	DOUT16

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```
[U14] Select output signal (Default I/O mode only)
  {output 13 – 16} (0: System, 1: User)
= 1   1   1   1
  {output 25 – 32} (0: System, 1: User)
= 0   0   0   0   0   0   0   0
```

When "Default (standard sequence)" is specified for [U11], outputs 13 ~ 16, 25 ~ 32 can be set as the user output signals or system output signals.

Specify the data in the following order.

= (output13) (output14) (output15) (output16)
(output25) (output26) (output27) (output28) (output29) (output30) (output 31)
(output32)

Output signal line	0: System	1: User
OUTPUT13	Reserved (for extending function)	DOUT13
OUTPUT14	Reserved (for extending function)	DOUT14
OUTPUT15	SV_RDY	DOUT15
OUTPUT16	BT_ALM	DOUT16
OUTPUT25	ACK	DOUT25
OUTPUT26	TEACH	DOUT26
OUTPUT27	EXTSIG	DOUT27
OUTPUT28	SYS_RDY	DOUT28
OUTPUT29	AUTORUN	DOUT29
OUTPUT30	ALARM	DOUT30
OUTPUT31	CYC_END	DOUT31
OUTPUT32	LOW_ST	DOUT32

[U15] Sequence Parameter

Set the values of internal relays that can be utilized as the contact inputs in the user's created sequence program.

```
[U15] Sequence Parameter (User I/O mode only)
= 0  0  0  0  0  0  0  0
```

Specify the data in the following order.
= (R510) (R511) (R512) (R513) (R514) (R515) (R516) (R517)

0 : OFF
1 : ON

This parameter is effective only when "User" is selected by parameter [U11].

[U16] Software limit

Specify the software limit values (JLIMT).

```
[U16] Software limit [deg] [mm]
  {+ Direction}
= 116.000  141.000  201.000  281.000  100.000  0.000
  {– Direction}
= –116.000  –141.000  –1.000  –281.000  –100.000  0.000
```

(+ Direction): Setting of plus (+) joint limit (+ soft stroke limit).
Values set in the joint limit setting change mode (JLIMT) are saved.

Set values:	=	116.000 (Axis 1)	121.000 (Axis 2)	201.000 (Axis 3)	281.000 (Axis 4)	0.000 (Axis 5)	0.000 (Not used)

(– Direction): Setting of minus (–) joint limit (– soft stroke limit).
Values set in the joint limit setting change mode (JLIMT) are saved.

Set values:	=	–116.000 (Axis 1)	–121.000 (Axis 2)	–1.000 (Axis 3)	–281.000 (Axis 4)	0.000 (Axis 5)	0.000 (Not used)

[U17] PLC alarm

Specify the PLC alarm message.

```
[U17] PLC alarm
= "8–269 PLCALM01"
= "8–270 PLCALM02"
= "8–271 PLCALM03"
= "8–272 PLCALM04"
= "4–077 PLCALM05"
= "4–078 PLCALM06"
= "4–079 PLCALM07"
= "4–080 PLCALM08"
= "1–037 PLCALM09"
= "1–038 PLCALM10"
= "1–039 PLCALM11"
= "1–040 PLCALM12"
= "1–041 PLCALM13"
= "1–042 PLCALM14"
= "1–043 PLCALM15"
= "1–044 PLCALM16"
```

The PLC alarm is an alarm which can be generated in the PLC when "User" is selected for "[U11] I/O mode". In this setting, message for each of such alarm can be specified.

Example 1: = "8–269 PLCALM01"

Specify an alarm message to be displayed in the area flanked by the quotation marks ("). Though up to thirty-two (32) characters can be specified, the leading six (6) characters are used to identify the alarm number, and a total of twenty-six (26) characters can be used for the message. DO NOT change the leading six (6) characters (alarm code) which is used to identify the alarm number.

[U18] Field bus

Specify the field bus. The field bus is an optional function.

[U18] Field bus
= –1 –1 0 0

= (Type) (Node) (Transmission rate) (Type of master)

(Type): Specify the type of the field bus.

–1 : Without field bus option (initial value)
1 : Profibus
37 : Device Net
144 : CC-LINK

(Node): Specify the node address of the field bus.
This address should not be identical with another device in the same network. The setting range differs with the type of the field bus.

–1 : Without field bus option (initial value)
0 ~ 127 : Profibus
0 ~ 63 : Device Net
1 ~ 64 : CC-LINK

(Transmission rate): Specify the transmission rate of DeviceNet. Make it coincide with the master transmission rate. Setting of Profibus is unnecessary.

Device Net
0 : 125 kbps
1 : 250 kbps
2 : 500 kbps

CC-LINK
0 : 156 kbps
1 : 625 kbps
2 : 2.5Mbps
3 : 5 Mbps
4 : 10 Mbps

(Type of master): Specify the type of the field bus master.
The bit string (endian) differs with the master. Make the value identical with the master.

0 : 16-bit big endian (DeviceNet made by OMRON)
1 : 16-bit little endian
2 : 32-bit big endian
3 : 32-bit little endian

Example: Set value = 37 4 1 0
DeviceNet, node 4, 256 kbps and master of OMRON-make are selected.

[U19] Password

The password is an optional function.

[U19] Password = “ “

= (Password MAX 8 character)

[U20] Conveyor Parameter

These parameters are exclusively used for the conveyor synchronization function. Each parameter consists of data in the two (2) columns. The data in the first column is used for CH1 and the data in the second column for CH2.

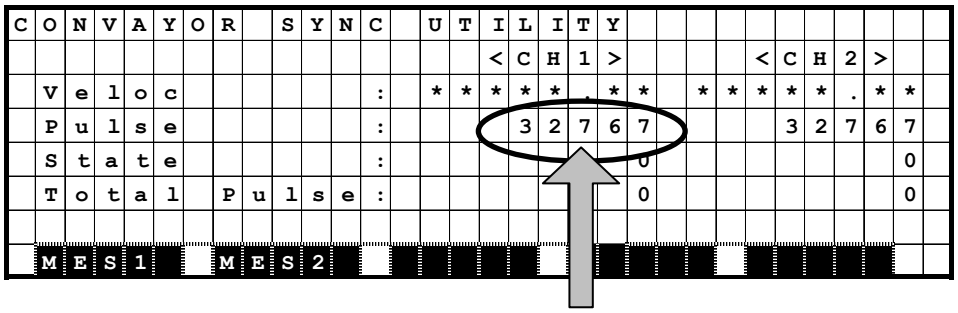
[U20] Conveyor Parameter
{Conveyor Parameter1(CH1,CH2)}
{Encoder rotated direction =0: A signal fast =1: B signal fast}
= 0 0
{Speed interpolation period [ms]}
= 3 3
{Conveyor timeout [ms]}
= 0 0
{Conveyor Sync Type =0:Linear, =1:ArcPositive, =2:ArcNegative}
= 0 0
{No use.}
= 0 0
{No use.}
= 0 0
{No use.}
= 0 0
{Conveyor Parameter2(CH1,CH2)}
{Moving distance per pulse [um/pulse] [mdeg/pulse]}
= 0.0 0.0
{ X center position }
= 0.0 0.0
{ Y center position }
= 0.0 0.0
{No use.}
= 0.0 0.0
{No use.}

= 0.0 0.0
{No use.}
= 0.0 0.0
{No use.}
= 0.0 0.0
{No use.}
= 0.0 0.0

{Encoder rotated direction} Encoder rotating direction

Data format : Integer type
Data setting range : 0 or 1
Example of value : 0

When the startup sequence of phases A and B is compared at pulse output of phases A and B from an encoder used, the pulse adding direction can be specified, taking phase A as the reference.
The value of this parameter is decided according to the encoder specifications used and the rotating direction of the encoder set to the conveyor.
Press the <UTILITY> key, then select the [CONV] menu. The following menu display appears.



Run the conveyor and specify the value in the direction in which the value of [Pulse] increases.

Example:
[1] When the value of [Pulse] increases while "0" is specified for the encoder rotating direction → Specify zero (0).
[2] When the value of [Pulse] decreases while "0" is specified for the encoder rotating direction → Specify number 1.

{Speed interpolation period [ms]} Encoder sampling period

Data format	:	Integer type
Data unit	:	msec
Data setting range	:	3 ~ 30000
Example of value	:	3

This parameter specifies the period (*) during which the counter value from the encoder attached to the conveyor is obtained.

* As the arithmetic operation period of the conveyor synchronization function is 3 ms, the actual parameter value is a multiple of this arithmetic operation period (3 ms) whose fractions are ignored.

[Cautions]
The conveyor synchronization function calculates the synchronization travel distance at minimum periods of 3 ms (default). It is also possible to specify a larger value, however. Determine the value, taking note of the following.

When the value is decreased (minimum value 3 msec):
- The trackability for the conveyor speed improves.
- If this value is too small, however, the synchronization speed changes even with conveyor pulsation, which may result in vibration of the robot.

When the value is increased:
- The synchronization speed is less affected by conveyor pulsation.
- The conveyor rotating angle and arithmetic operation error can be absorbed.
- The trackability for the conveyor speed deteriorates.

Unless the pulsation is caused, specify the default value (3 msec).

Example:
When the conveyor sampling period is set to 15 msec, the synchronization travel distance is figured out based on the previous conveyor sampling period average speed (Vave).

{Conveyor timeout [ms]} Conveyor stop judgment time

This parameter specifies the conveyor stop judgment time. Unless the encoder counter value changes during the time set by this parameter value, the system interprets that the conveyor is stopped, and outputs the error of “001–048: Conveyor Stop”. As this is the error of level 1, programmed operation continues.

Data format : Integer type
Data unit : msec
Data setting range : 0 ~ 30000
Example of value : 3000

- Normally, the conveyor stop time is specified by the 2nd argument of the SYNC command. Unless the 2nd argument is specified, the conveyor stop is judged with this parameter value.
- When the 2nd argument of the SYNC command is not specified and this parameter value is zero (0), the system will not judge the conveyor stop.

{Conveyor Sync Type =0: Linear, =1 ArcPositive, =2: ArcNegative}
Type of conveyor synchronization

Data format : Integer type
Data setting range : 0 ~ 2
Example of value : 0

This parameter specifies the type of the conveyor synchronization function. Select a desired type from the following four (4) types.

0 : Linear conveyor
1 : Arc conveyor (forward)
2 : Arc conveyor (reverse)

For the rotating direction in the arc conveyor synchronization, the right-hand screw direction of the conveyor rotating axis (*) is taken as "forward".

* The conveyor rotating axis signifies a vector parallel with the Z-axis in the world coordinate system, which passes the center position of arc conveyor rotation.

{Dummy data} **Reserved area**

Data format : Integer type

Area reserved for the system.

{Moving distance per pulse [μm/pulse] [mdeg/pulse]}
Travel distance per pulse of encoder

This parameter specifies the conveyor travel distance (μm) per one (1) pulse of the encoder set to the conveyor.

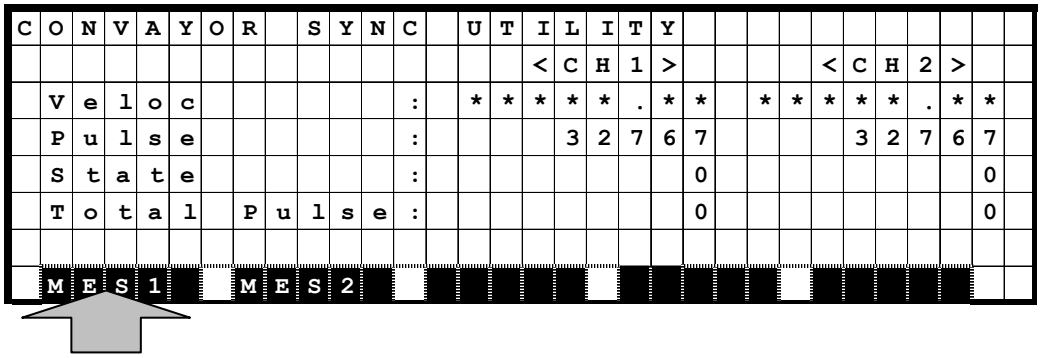
Data format : Real number type
Data unit : μm
Example of value : 0

If the total number of encoder pulses is ΔP (pulse) when the conveyor moves by ΔL (mm), the conveyor travel distance per one (1) pulse of the encoder α (μm) can be calculated from the following equation.

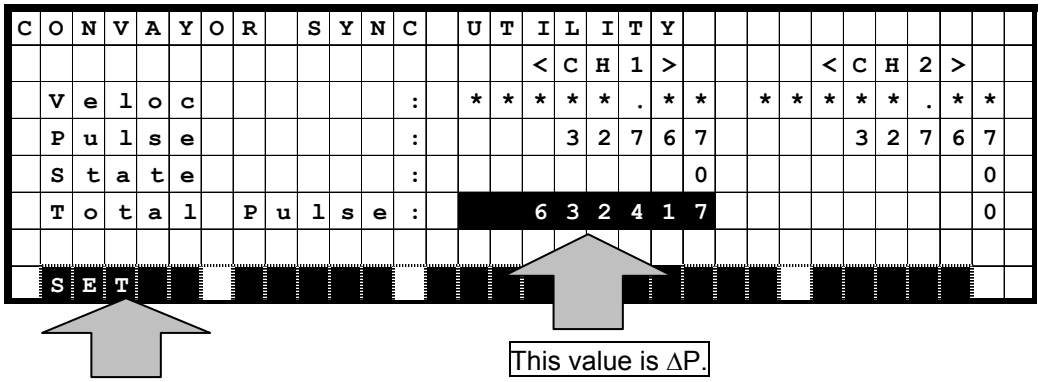
$\alpha = (\Delta L / \Delta P) * 1000$

The total number of encoder pulses (ΔP) can be obtained in the following manner.

(1) Press the <UTILITY> key and select the [CONV] menu. Then the following menu appears on the display.



(2) To count the number of pulses of CH1 or CH2, select [MES1] or [MES2], respectively. After moving the conveyor by ΔL, press the <SET> key.



(3) When you press the <SET> key, the present total number of pulses (Total Pulse) is shown on the display. Then figure out the travel distance per one (1) pulse (α μm) by substituting this value for "ΔP" in the above equation. (To measure the value again, press the [MES1] or [MES2] key.)

{X center position}, {Y center position}
Center position X, Y (world coordinate system) of arc conveyor rotation

This parameter specifies the rotation center position of the arc conveyor in units of mm, when seen from the origin of the world coordinate system.

Data format : Real number type
Data unit : mm
Example of value : 0

{Dummy data} Reserved area

Data format : Real number type

* Area reserved for the system.

[U21] Accel limit function

When this parameter is made valid, acceleration/deceleration is limited according to the Z-axis height. This function is effective only for the SCARA robot system.

Parameter format

[U21] Accel limit function = 0

[U22] Master mode

It is possible to specify the master mode effective after power ON by setting this parameter beforehand. Select and program either of the following character strings given below.

"TEACHING"	:	Teaching
"INTERNAL"	:	Internal auto
"EXT.SIGNAL"	:	External auto (External signal control)
"EXT.HOST"	:	External auto (Host computer control)
Other than the above	:	Master mode as backed up at power OFF or condition selected by the panel key.

* If there is difference in setting from the master mode key switch, priority is given to the setting of the same switch. (TS2000)
* When the external auto mode is selected initially after the system has started up in the TEACHING or INTERNAL mode, priority is given to the setting of this parameter. (Changeover of external signal control/host computer control in TS2000)

Parameter format

```
[U22] Master mode
= "EXT.SIGNAL"
```

[U23] PLC communication register address

This parameter is exclusively used for the die-cast unloading robot. Communication is done with the TC200 made by Toshiba Machine.

Current position	:	Address where current coordinates of the robot (in the work coordinate system) are to be sent.
Alarm	:	Address where alarm information is to be sent.
Master Mode	:	Address where master mode information is to be sent.
Alarm Resetting	:	Address where alarm reset from the PLC is to be monitored.
spare	:	Reserved.
spare	:	Reserved.

Parameter format

```
[U23] PLC communication Register address
= {Current Position/ Alarm/ Master Mode/ Alarm Resetting/ spare /spare}
=  4  2560  2580  0  0
```

[U24] Serial data error check

This parameter is used as the ON/OFF changeover flag of error check for non-procedural communication and simple procedural communication. Specify zero (0) normally.

Parameter format

```
[U24] Serial data error check
= {0: checked, 1: passed}
= 0
```

[U25] Common data & program file function setup

This parameter serves as the ON/OFF flag of the common program and common data functions in the SCOL language specifications.

0 : Invalid
1 : Common data (DATA. TBL)
2 : Common program (SCOL. PRG)

Parameter format

```
[U25] Common data & program file function setup
= {0: Disable, 1: DATA. TBL, 2: SCOL. PRG}
= 0
```

[U26] Bypass function parameter

In the M–TO (move to teach point) function, as the robot arm moves directly to the teach point, it may collide with a workpiece present midway.
When this bypass (move to teach point by bypassing) function is made effective, the robot once retracts its hand, moves to just before the target position, then to the target position. This function is used only for the SCARA robot system and LCDR robot system.

<SCARA robot>

1. Moves by retracting from current position P0 to P0a.
2. "Parallel move + rotation" from position P0a to P1a.
3. Moves by extending from position P1a to P1.

<LCDR robot>

1. Leaves a workpiece.
2. Moves by retracting from current position P0 to P0a.
3. "Parallel move + rotation" from position P0a to P1a.
4. Moves by extending from position P1a to P1.
5. Contacts the workpiece.

(1) Bypass function ON/OFF setting
= 2 (0: OFF, 1: M–TO & bypass ON, 2: Only bypass ON)

(2) Retract position parameter (Specify the absolute position rather than the relative travel distance.)
= 0.0 0.0 200.0 0.0 0.0 0.0 (For SCARA robot: Only Z axis is effective.)
↑ Z-axis
= 100.0 100.0 0.0 0.0 0.0 0.0
↑ RR-axis ↑ RL-axis
(For LCDR robot: Only RR axis and RL axis are effective.)

(3) Z-axis offset parameter for LCDR
= –10.0 –10.0

Parameter format

[U26] Bypass Function Parameter
{Function Enable SW (0: Disable 1: M–to & Bypass 2: Bypass)}
= 0

{Retract Position}
= 100.0 100.0 0.0 0.0 0.0 0.0

{Z Axis Offset Parameter} ← Exclusive for LCDR.
= –10.0 –10.0

[U27] Speed limit Function

When this function is made valid, motion speed of the robot is controlled in such a manner that it will not exceed the maximum speed.

Parameter format

```
[U27] Speed limit Function
{0: Disable, 1: Enable}
= 1
```

[U28] Vision + Conveyor Tracking Parameter (CH1, CH2) (Only for TS2000 SCARA system)

The "vision + conveyor synchronization" function which uses this parameter is the optional function for the TS2000 SCARA system.
For the setting of this parameter, see the "Vision sensor + conveyor" synchronization function manual (STE 73912) provided separately.

[U29] Function select switch

This parameter selects ON or OFF of the controller functions.
The functions which can be effective are assigned to each element.

1 : Function ON
0 : Function OFF

Parameter format

```
[U29] Function select switch
= 0  0  0  0  0  0
= 0  0  0  0  0  0
= 0  0  0  0  0  0
= 0  0  0  0  0  0
```

The effective functions of each element are described below.

First line (from left)

1st function: Input character string conversion function

If a character string other than the numerals has reached in the input processing, an error occurs. To avoid this, any character string reached other than the numerals can be input as zero (0).

2nd function: "Move to teach point" function (MOV–TO)

A motion command in the "move to teach point" mode can be changed over according to the current guide coordinates. The same is also applicable to the BYPASS function.

Guide coordinates	Motion command
JOINT	MOVE
TOOL WORK WORLD	MOVES

(1) Function ON/OFF

The 2nd function of function select switch [U29] of the user parameter (USER. PAR) is used to select ON or OFF of this function.

(2) Current position display during "move to teach point" mode

At the same time that the "move to teach point" mode has been selected, current position is displayed in the world coordinate system.

When the robot has reached the teach point, or when the ESC key has been pressed, the previous data editing mode takes effect again.

3rd function: Keeping of latch register

When the user parameter is specified as shown below, the latch register was not previously kept in the same way as in "3: USER BACKYP RAM" in the USER FLASH mode.

[U11] 1/OMODE {DEFAULT/USER} (0: DEFAULT, 1: USER RAM, 2: USER FLASH, 3: USER BACKUP RAM)
= 2

To settle this problem, the mode including the register keeping function is added to select ON or OFF of the latch register keeping by means of parameter setting.

0 : Not kept (as in the past)
1 : Kept

* When the other mode is selected for [U11] I/OMODE:

0: DEFAULT	Not kept
1: USER RAM	Not kept
2: USER BACKUP RAM	Kept

4th function: Prohibition of I/O change in EXT mode
I/O change on the screen called by "UTILITY → SIGNAL" or "UTILITY → AUX" is prohibited in a mode other than the MANUAL mode.

5th function: Extending the numbers of Profibus I/Os to 128/128
The numbers of the Prohibus I/Os can be increased from 64/64 to 128/128.
This function is effective only for the Profibus.

Other functions: Reserved for the system.

EXTRNSEL.SYS (TS2000)

Specify the program file corresponding to the file select signal line.

```
Copyright(C) 2001 by TOSHIBA MACHINE CO.,LTD.
All rights reserved.
External select file   "EXTRNSEL.SYS"
*** [ 00 – 0F ] *****
= "PROG00"
= "PROG01"
= "PROG02"
= "PROG03"
= "PROG04"
= "PROG05"
= "PROG06"
= "PROG07"
= "PROG08"
= "PROG09"
= "PROG0A"
= "PROG0B"
= "PROG0C"
= "PROG0D"
= "PROG0E"
= "PROG0F"
*** [ 10 – 1F ] *****
= "PROG10"
= "PROG11"
= "PROG12"
= "PROG13"
```

The number of effective files differs with the bit length according to the setting of [U07], and the files are assigned to the file select signals in turn, starting with the top file.

Bit length 1:	0	~	1 (binary number)	Two (2) files
Bit length 2:	00	~	11 (binary number)	Four (4) files
Bit length 3:	000	~	111 (binary number)	Eight (8) files
:			:	
:			:	
Bit length 8:	00000000	~	11111111 (binary notation)	256 files

Specify the file name following the equal (=) code.
Unless the equal (=) code is specified at the top of the line, it is processed as a comment.

Example: = "PROG1"
PROG1 is assigned to the file select signal.

APPROVED BY:

CHECKED BY:

PREPARED BY: